

Yifan Feng

Ph.D. Candidate in Entomology — mosquito genomics, chromosomal inversions, and cytogenetics

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Education

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| PhD Virginia Polytechnic Institute and State University , Entomology <ul style="list-style-type: none"> • Advisor: Dr. Maria V. Sharakhova • Dissertation: <i>Chromosomal Inversions Differentiate Mosquitoes in the <i>Culex pipiens</i> Complex</i> • Department of Entomology and Fralin Life Sciences Institute • Coursework: Molecular Genetics, Bioinformatics, Genomics, Biochemistry, Experimental Design | Blacksburg, VA
Aug 2021 – Dec 2026 |
| BS Virginia Polytechnic Institute and State University , Biological Sciences | Blacksburg, VA
Aug 2017 – May 2021 |
| BS Virginia Polytechnic Institute and State University , Food Science and Technology | Blacksburg, VA
Aug 2017 – May 2021 |

Research Experience

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| Virginia Tech — Department of Entomology & Fralin Life Sciences Institute , Graduate Research Assistant <ul style="list-style-type: none"> • Led multi-species genomic projects integrating Hi-C scaffolding, Nanopore long-read sequencing, and FISH cytogenetic mapping to assemble chromosome-scale mosquito genomes. • Designed and validated PCR breakpoint assays to genotype chromosomal inversions across natural populations, linking structural variants to climatic adaptation and vector traits. • Conducted population genomic analyses (FST, PCA, AMOVA) to quantify differentiation and gene flow among <i>Culex pipiens</i> populations across North America. • Performed molecular dissections, DNA extraction, library preparation, and fluorescence microscopy imaging; maintained mosquito colonies. • Mentored two undergraduate researchers in PCR optimization, experimental design, and scientific writing. • Coordinated with collaborating groups across nine countries on the <i>Aedes aegypti</i> inversion study published in <i>Genome Biology and Evolution</i>. | Blacksburg, VA
Aug 2021 – present |
| Virginia Tech — Department of Food Science and Technology , Undergraduate Researcher <p>Aerosolized <i>Listeria</i> contamination from cavitation treatment of raw produce.</p> <ul style="list-style-type: none"> • Applied microbubble cavitation treatments and quantified resulting aerosolized bacterial contamination. • Built MATLAB models of cavitation effects and ran predictive simulations on university high-performance computing resources. • Compiled, visualized, and statistically analyzed experimental results to identify contamination trends. | Blacksburg, VA
Sept 2020 – Dec 2020 |

Molecular Genetics Laboratory, Research Intern

- Performed DNA extraction, PCR amplification, and sequencing for mosquito population genetics studies.
- Managed sample inventory and quality control for field and laboratory collections.
- Gained hands-on experience with genome assembly workflows and variant calling pipelines.

Blacksburg, VA

Mar 2020 – Dec 2021

Publications

Chromosomal Inversions and Their Potential Impact on the Evolution of Arboviral Vector *Aedes aegypti*

July 2025

Jiangtao Liang, Noah H. Rose, Ilya I. Brusentsov, Varvara Lukyanchikova, Dmitriy A. Karagodin, Yifan Feng, et al.

[10.1093/gbe/evaf118](https://doi.org/10.1093/gbe/evaf118) (Genome Biology and Evolution 17(7), evaf118)

Hi-C Proximity Ligation Approach Identified Chromosomal Rearrangements in *Culex pipiens* Mosquitoes

2025

Yifan Feng, Varvara Lukyanchikova, Jiangtao Liang, Dmitriy A. Karagodin, Ilya I. Brusentsov, Megan L. Fritz, et al.

(American Journal of Tropical Medicine and Hygiene 112(6) — ASTMH Annual Meeting abstract)

Chromosomal Rearrangements Differentiate Species of the *Culex pipiens* Complex

2025

Maria V. Sharakhova, Yifan Feng, Varvara Lukyanchikova, Jiangtao Liang, Dmitriy A. Karagodin, et al.

(Conference proceedings)

Chromosomal Inversions Differentiate Mosquitoes in the *Culex pipiens* Complex

2026

Yifan Feng, Varvara Lukyanchikova, Jiangtao Liang, Ilya I. Brusentsov, Dmitriy A. Karagodin, Megan L. Fritz, Maria V. Sharakhova

(Manuscript in preparation)

Chromosome-Scale Genome Assembly of *Aedes albopictus*

2026

Yifan Feng, et al.

(Manuscript in preparation)

Conference Presentations

- **Talk (accepted)** — *Hi-C proximity ligation reveals chromosomal inversions potentially associated with host preference in Culex pipiens mosquitoes*. Student 10-Minute Presentation Competition, Entomology 2026 (ESA Annual Meeting), Columbus, OH, November 2026.
- **Poster** — *Hi-C based discovery of structural variants in the Culex pipiens complex*. American Society of Tropical Medicine and Hygiene (ASTMH) Annual Meeting, November 2024.
- **Presentation** — *Chromosomal inversions differentiate mosquitoes in the Culex pipiens complex*. CeZAP Infectious Disease Symposium, Blacksburg, VA, October 2023.

Teaching and Mentoring

Virginia Tech — Department of Entomology, Graduate Teaching Assistant, ENT 2004 (Insects and Human Society)

Blacksburg, VA

Aug 2023 – May 2024

- Led weekly laboratory sessions and discussion sections for a general-education course serving non-biology majors.
- Taught insect taxonomy, physiology, and ecology exercises; graded laboratory reports and examinations.
- Developed visual teaching materials and provided one-on-one mentorship.

Sharakhova Lab, Virginia Tech, Research Mentor, Undergraduate Researchers

- Mentored two undergraduate researchers through full project cycles — PCR optimization and troubleshooting, experimental design, data interpretation, and scientific writing.

Blacksburg, VA

Jan 2024 – Dec 2025

Technical Skills

Molecular Biology: DNA/RNA extraction, PCR, qPCR, RT-qPCR, ddPCR, gel electrophoresis, cloning, nucleic acid purification

Library Prep & Sequencing: Hi-C library preparation, Illumina library prep, Sanger sequencing, Nanopore long-read workflows

Cytogenetics & Imaging: Chromosome preparation, FISH probe synthesis and hybridization, fluorescence microscopy (Zeiss ZEN), ImageJ

Genomics & Bioinformatics: Hi-C scaffolding (Juicer, Juicebox), genome assembly, trio-binning, gene annotation, variant calling, alignment (BWA, Minimap2)

Population Genetics: FST, PCA, AMOVA, STRUCTURE, PLINK, inversion genotyping

Cell & Assay Work: Mammalian and insect cell culture, sterile technique, ELISA, Western blot, BSL-2 practice

Programming & Software: R (ggplot2), Python, Linux/Unix, MATLAB, Galaxy, Geneious, BLAST/NCBI, GraphPad Prism, BioRender

Scientific Communication: Manuscript preparation, grant proposal writing, poster and oral presentation, mentoring, laboratory documentation

Service and Professional Membership

- Volunteer, Blacksburg Mosquito Control Initiative — field collection and public outreach on vector-borne disease prevention.
- Member, Entomological Society of America.
- Member, Virginia Mosquito Control Association.